



Load digitized handwriting

PaHaW: 75 participants / 597 recordings / 8 tasks
Position, time, contact, pressure and pen angles

Audit records and define participant folds

5 repeats × 5 outer folds; common across tasks

Render the next encoding

Static → 4 single signals → 4 RGB triplets

Extract LPQ; tune and evaluate SVM

Tune in 3 inner folds, refit, then predict outer test
Repeat for every task and outer fold

[yes] next encoding

More encodings?

[no]

Compare all nine encodings; retain SAZ

Speed (R), altitude (G), azimuth (B)
Exploratory selection using outer scores

Evaluate models on LPQ

SVM predictions already available
Tune and evaluate LR and RF

Train a CNN on RGB pixels

Select epochs on inner holdout
Refit and evaluate outer test

Compare methods and combine tasks

Task-level scores; participant-level majority vote

Explain held-out CNN predictions

Grad-CAM, occlusion and perturbation checks

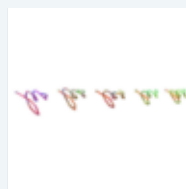
Inspect classification and explanation results

SVM: participant classification

	H	PD
H	58.9%	41.1%
PD	28.6%	71.4%

Rows: observed
Columns: predicted

CNN input



Grad-CAM

